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A paleomagnetic study of the Mull lava succession

James M. Hall* *Department of Geology, Dalhousie University, Halifax, Nova Scotia, Canada*

R. L. Wilson and Peter Dagley *Sub-department of Geophysics, University of Liverpool, Liverpool L69 3BX*

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Summary. A paleomagnetic study has been made of a succession of 139 non-overlapping basaltic lavas, representing 91 per cent of the longest remaining succession of flows associated with the Paleogene Mull volcano. All the lavas have experienced considerable hydrothermal alteration, probably at up to several million years after initial magnetization and frequently with alteration to the opaque minerals and the production of new potentially magnetic phases. The question of whether directional remagnetization has taken place while preserving within-unit directional consistency and discreteness of unit mean direction is discussed. Extensive directional remagnetization is excluded as an explanation for the data. If stable directions obtained by alternating field remanence cleaning coincide with original TRM directions then a mean pole position for all temporally independent lava directions from the British Tertiary igneous province is at 71.9° N, 167.2° E, with k: 22 and α_{95}: 3.0°. This pole is significantly different from the geographic pole. If the difference in palaeomagnetic and geographic poles is interpreted in terms of absolute plate motion, then 2010 km of northwards motion of the western part of the Eurasian Plate, at 3.7 cm/yr, has taken place over the last 55 Myr. This motion has implications for the geological history of the Arctic and for the complexity of mantle motions.

1 Introduction

This paper forms one of a series describing a major paleomagnetic re-examination of the British Tertiary igneous province. This volcanic province was one of the first studied and is described in a series of classic papers and memoirs published during the late nineteenth and early twentieth centuries. The province also provided materials for a number of the early experiments in paleomagnetism (Bruckshaw & Robertson 1949; Bruckshaw & Vincenz 1954;

* Present address: University of California, Scripps Institution of Oceanography, Deep Sea Drilling Project A-031, La Jolla, California 92093, USA.

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Vincenz 1954; Leng 1955). Some of the important results from this re-examination are the presence of many groups of volcanic units with closely similar directions, representing either a high frequency of volcanic activity or periods of low secular variation, and the widespread occurrence of volcanic units with southerly directed, shallow remanence inclinations. The results of related studies on material collected for paleomagnetic work include a number of correlations between features of the opaque minerals and remanence polarity in Mull dykes (Ade-Hall & Wilson 1969), the recognition of the variable effect of regional hydrothermal alteration in the province (Ade-Hall, Palmer & Hubbard 1971), the development of new techniques for paleointensity determination (Allen 1970), the study of within-lava variation of magnetic and opaque mineralogical properties in an Antrim tholeiite (Lawley & Ade-Hall 1971) and a series of K-Ar age studies (Purdy *et al.* 1973; Mussett *et al.* 1973).

2 Geological setting

During the nineteenth century geological exploration of Britain, the Isle of Mull was recognized to consist largely of the eroded remanent of a major volcanic centre (Judd 1889) and since then has been the site of external volcanological mapping and research culminating with the publication of the classic memoir by Bailey *et al.* (1924). More recently Walker (1970) has, by comparison with similar centres in Iceland, estimated that the original maximum thickness of the lava pile exceeded 2000 m. Today only the lowermost 1200 m of lavas are preserved. This lower part of the original lava pile has been subjected to extensive hydrothermal alteration (Walker 1970; Ade-Hall *et al.* 1971). Regional alteration followed the heating of ground water by the geothermal flux with locally more intense effects in an annular zone surrounding the central region of major granophyric and gabbroic intrusion.

Geological studies of the floor of the Atlantic during the last two decades encourage attempts to answer the question of the overall setting of the Mull volcano and the other features of the British Tertiary igneous province. The age of the province is largely between 50 Myr and 60 Myr, an interval that was marked by major changes in the spreading pattern in the North Atlantic (Vogt *et al.* 1969; Le Pichon, Hyndman & Pautot 1971; Laughton 1971; Pitman & Talwani 1972). Prior to 60–63 Myr, opening to the north of the Gibbs Fracture Zone was largely between Greenland and North America. At about 63 Myr ago a triple junction developed to the south of Greenland and spreading took place to both east and west of Greenland until about 49 Myr. At this latter time spreading to the west of Greenland was effectively terminated while to the north of Iceland the spreading axis moved westwards from a line now marked by seamounts to the Jan Mayen Ridge. Since the interval during which the British Tertiary igneous province was active both begins and ends with changes in the spreading patterns in the North Atlantic it is reasonable to try to fit the activity into this larger framework. One possibility is that the British Tertiary province centres represent the failed arm (Burke & Dewey 1972) of a triple junction active in the vicinity of the Faeroes during the 63–49 Myr interval, with one of the temporarily successful arms being responsible for the Rockall Trough. The ages of the Faeroes basalts (Lower Eocene – Laufeld 1965; 49–65 Myr – Tarling & Gale 1968) are consistent with this model. From this model the average rate of crustal extension over the 50–60-Myr interval resulting from the intrusion of the dyke swarms of the British Tertiary igneous province is 0.02 cm/yr, or two orders of magnitude less than typical rates on the successful arms of a triple junction. Alternative models have been suggested by Dunham (1970) who proposed that activity at the then adjacent Reykjanes Ridge was responsible for the British Tertiary province activity and by Duncan, Peterson & Hargreaves (1972) who proposed a thermal plume or hot-spot trace (Wilson 1963; Morgan 1971, 1972a) origin. However, large (17 cm/yr) and thus probably unreasonable plate velocities are required for this latter explana-

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tion to be correct. There is also conflict between paleomagnetic data and the polar wander curve predicted by the supposed plume trace.

Table 1. Mull lava section. Location of components of composite section (Reference UK Ordnance Survey, 1 inch to 1 mile (7th Series) sheet 45, Sound of Mull) (See also Fig. 1)

Component	Location
Flows A03C001–7	Low-lying area at Gribun between NM449341 and NM453341
C400	At foot of cliff directly adjacent to road B8035. NM474363
C015–26	Prominent nose above Rubh a' Ghearrain. NM458360
C030–65	In gully of un-named stream between NM456355 and NM462348
C070–125	Abhainn Dhiseig stream gully from NM499350 to NM517338
C150–165	Largely north side of ridge between Abhainn Dhiseig and Coire Coran (Coire nam Fuaran) NM515336 to NM522333
C167–200	Stream head in Coir' Odhar to Ben More summit. NM527332 to NM528330

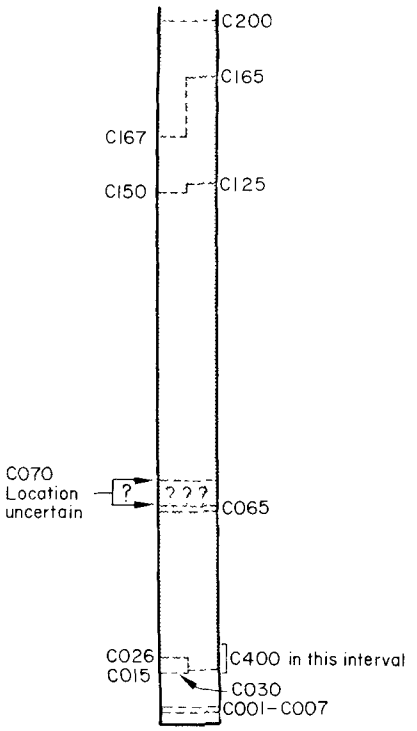


Figure 1. Composite stratigraphic section for the part of the Mull lava pile described in this paper.

3 Sampling procedure

Specimens were 2.5-cm diameter drill cores obtained using portable diamond drills based on the design of the US Geological Survey (Doell & Cox 1967). With rare exceptions, where flows consisted of highly friable material, four geographically oriented cores were obtained from each rock unit.

No single section contains the full number of remaining flows and as a result of this our profile comprises a composite of seven different sections listed in Table 1. Using dip and strike measurements made in the field these seven sections have been fitted together (Fig. 1). The complete profile consists of 139 non-overlapping flows, the lowest resting on

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Table 2. Mean directions of cleaned natural remanence for the Mull lavas. Where an average direction is given for several units, it is assumed that only one temporally independent field direction is involved, directions for the individual units are given in the right-hand column. Flows are listed in stratigraphic order with A03C001 at the base of the section.

Unit number	D	I	k	α_{95}	$\frac{N \text{ used}}{N \text{ sampled}}$	
A03C001 to 6	220	-71	215	3.9	21/25	205, -70; 237, -66; 229, -67; 212, -75; 219, -71; 211, -74.
C007	178	-73	204	3.9	4/4	
						No flows with code numbers C008-14.
C015-24, 31-33	162	-62	487	1.8	43/45	158, -63; 164, -62; 157, -63; 164, -60; 165, -61; 164, -64; C021:R; 154, -60; 172, -63; 173, -65; 152, -62; 168, -60; 154, -61.
C025	190	-61	247	3.5	4/4	
C026	200	-53	94	5.9	3/4	
C030	164	-55	194	4.0	4/4	No flows with code numbers C027-29.
C034	178	-67	2000	1.1	5/5	C031-33 included with C015-24.
C035	238	-65	159	5.6	2/3	
C036	208	-61	516	2.5	3/4	
C037/8	186	-48	278	6.0	7/8	189, -45; 182, -50.
C039	147	-68	151	4.5	4/4	
C040	R†					
C041	155	-58	299	3.2	4/4	
C042	177	-62	181	4.3	3/4	
C043	192	-56	201	4.0	3/4	
C045	212	-68	288	3.4	3/4	C044 very friable; not sampled.
C046/7	148	-54	361	5.2	6/6	148, -51; 147, -57.
C048/9	137	-45	958	3.2	6/6	135, -44; 140, -45.
C050	R					
C051	139	-62	1097	1.7	4/4	
C052	167	-62	537	2.4	4/4	
C053, 53A-53D/54	149	-60	1175	1.7	18/19	146, -62; 149, -60; 146, -61; 153, -60; 150, -62; 152, -57.
C055	R					
C056	207	-26	349	3.0	4/4	
C057	214	-34	96	5.7	4/4	
C058	206	-43	980	1.8	4/4	
C059	R					
C060	160	-42	347	3.0	4/4	
C061-4	210	-58	426	3.4	15/17	200, -60; 209, -58; 215, -58; 214, -56.
C065	Very scattered directions.					Flow on hilltop, possible case of remagnetization by lightning strike.
C070	206	-66	213	3.8	4/4	No flows with code numbers C066-69.
C071	196	-56	246	3.5	4/4	
C072	181	-67	397	2.9	3/4	
C073	181	-57	93	5.9	3/4	
C074	175	-51	165	4.3	4/4	
C075	157	-58	86	6.0	4/4	
C076	206	-67	304	3.2	4/4	
C077	Very scattered directions.					Possibly remagnetized by lightning strike.

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Table 2 – *continued*

Unit number	*D*	*I*	*k*	α_{95}	*N* used / *N* sampled	
C078	173	-65	102	5.5	4/4	
C079	225	-60	235	3.6	4/4	
C081-88	185	-37	359	2.6	30/33	C080 assigned to a sheet. C081-99: 185, -41; 181, -35; 184, -36; 186, -34; 181, -35; 186, -37; 191, -43; 190, -37.
C089/90	193	-58	709	3.7	8/8	194, -56; 191, -60.
C091	175	-46	186	4.1	4/4	
C092	193	-63	248	3.5	4/4	
C093	187	-52	585	2.1	4/4	
C094/5	179	-53	518	4.4	8/8	179, -50; 178, -55.
C096	R					
C098	R					C097 assigned to a sheet.
C099/100	183	-45	1381	2.7	7/8	182, -46; 183, -43.
C101	183	-54	559	2.3	4/4	
C102	176	-43	271	3.4	4/4	
C103	176	-47	94	5.7	4/4	
C104	191	-63	1076	1.7	3/4	
C105/6	169	-42	1464	2.6	7/8	171, -43; 168, -41.
C107	173	-52	518	2.4	4/4	
C108	176	-38	156	4.0	4/4	
C109/110	185	-52	2166	2.1	7/8	187, -52; 183, -52.
C111	172	-60	578	2.3	4/4	
††C112/3	186	-58	325	5.5	7/8	184, -61; 188, -55.
C114	163	-24	1480	1.4	4/4	
C116	190	-61	712	2.1	4/4	C115 assigned to a sheet.
C117-25	186	-50	206	3.5	27/32	183, -57; 186, -52; 191, -46; 192, -48; 186, -53; 183, -41; 184, -47; C124: R, 185, -54.
C150	156	-45	137	4.7	4/4	Ben More Mugearite. No flows with code numbers C126-149.
C151	175	-54	210	3.8	4/4	
C152/3	194	-63	2723	1.9	7/8	193, -62; 195, -64.
C154	184	-57	610	2.2	4/4	
C155/6	170	-54	1404	2.6	6/8	170, -52; 171, -55.
C157	One specimen only.					
C158	Very scattered.					Probably remagnetized by lightning strike.
C159	Very scattered.					Probably remagnetized by lightning strike.
C160	R					
C161/168	182	-81	183	7.3	7/7	Stratigraphic equivalents. 188, -77; 168, -85.
C162/169	182	-62	227	6.6	8/8	Stratigraphic equivalents. 177, -65; 187, -59.
C163/170	170	-54	896	3.3	6/7	Stratigraphic equivalents. 172, -55; 168, -52.
C164/171/2	179	-58	271	4.9	10/11	C164 stratigraphic equivalent of C174/5.
C165/174/5	187	-67	733	3.0	11/12	C165 stratigraphic equivalent of C174/5. 183, -68; 186, -69; 190, -64. No flow with code C166.
C176	183	-53	298	3.3	3/4	C173 assigned to a sheet.
C177-183	180	-63	479	2.4	7/8	180, -63; 172, -66; 191, -67; 181, -64; 176, -63; 178, -61; 180, -59.

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Table 2 – *continued*

Unit number	D	I	k	α_{95}	N used / N sampled	
C184/5	207	-63	1801	2.3	6/8	209, -64; 205, -62.
C186-9	185	-55	960	2.3	13/16	184, -54; 182, -53; 186, -54; 190, -57.
C190/1	185	-47	238	6.4	7/8	181, -44; 189, -49.
C192	177	-55	480	4.2	4/4	
C193	195	-58	139	7.8	2/2	
C194	179	-66	175	4.3	3/4	
C195	155	-57	415	2.7	4/4	
C196	R					
C197	176	-57	109	5.5	3/4	
C198	206	-53	121	4.5	4/5	
C199	188	-52	134	4.9	3/4	
C200	201	-47	131	6.1	2/4	

Notes:

† Implies clearly a reverse polarity, but statistics of mean too poor to quote a numerical value.
†† Thermal demagnetization results for a sample from this flow are discussed in the text.

Tertiary volcanic sediments and the uppermost forming the peak of Ben More. The total thickness of flows is 1200 m and of this we have sampled close to 1090 m, the remaining 110 m (9.2 per cent) is obscured by landslips at the foot of the lava escarpment in the vicinity of Gribun. Checks on the geological continuity of the sections were made both in the field and by examination of aerial photographs of the area (V:CPE/SCOT/352: 17APR48, nos 5021/3/5, 5046/8).

4 Natural remanence

4.1 MEASUREMENTS

Natural remanence (NRM) measurements were made with a parastatic magnetometer. The NRM of each specimen was partly demagnetized by the alternating field method in 5 mT (50 Oe) steps to a peak field 20 mT. Further steps were carried out where routine demagnetization failed to produce a stable direction or where within-flow directional agreement was not achieved. Continuous thermal demagnetization was carried out for a single sample, C112.14.

4.2 FLOW MEAN REMANENCE DIRECTIONS

The first stage in NRM direction analysis was to obtain flow mean directions (Table 2) wherever the requirement for original thermoremanence of within-unit directional consistency was satisfied (Doell & Cox 1963). The averaging procedure utilized a computer search for the minimum scatter of directions for specimens from each unit (Dagley & Ade-Hall 1970), followed by a reassessment of the data from units yielding an initial α_{95} value in excess of 5°. For these units minor corrections to the data or rejection of one or, at the most, two specimens usually brought the α_{95} value for the modified mean direction to within the 5° limit. Grounds for rejection included insufficient alternating field cleaning, often for lightning-strike remagnetized material, exceedingly soft remanence or inferred field or laboratory error. In all 139 (89 per cent) of the 156 flows yielded statistically acceptable mean directions. For the remaining flows it was usually possible to quote the polarity.

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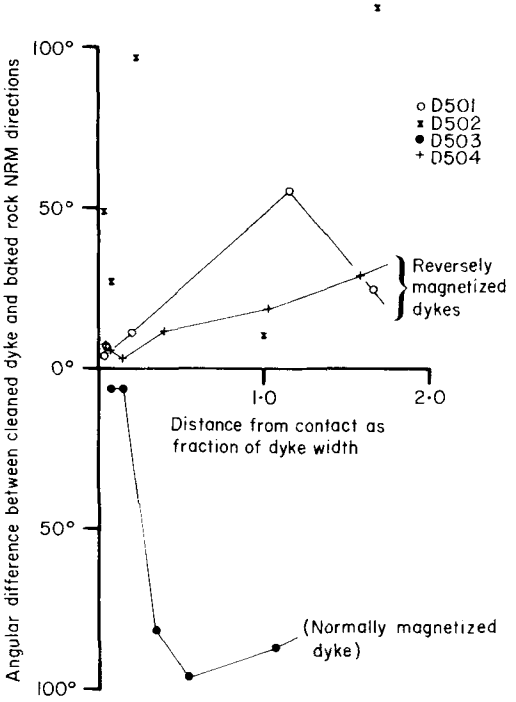


Figure 2. Divergence between dyke and baked rock cleaned NRM directions for dykes D501–504.

5 Originality of cleaned NRM directions

Before discussing the paleomagnetic results the problem must be considered of how well the cleaned NRM directions represent ambient field directions at the time of the initial cooling of flows. This is a very important question in view of the high degree of alteration of the flows. Flows C001 to C065 and C400 have experienced regional hydrothermal alteration in laumontite zone conditions (170–300°C reheating in the presence of abundant groundwater) and flows C070–C200 in prehnite or epidote zone conditions ($\geq$ 300°C reheating in the presence of abundant groundwater) (Walker 1970). Alteration at up to several million years after eruption, which produces extensive visible changes, including phase splitting, in the iron-titanium oxide minerals of the flows, must be suspected of imprinting new remanence directions on original cooling thermoremanence directions. Evidence bearing on this question from the Mull lava results is of three kinds; field relationships, remanence analyses and analogies with other studies.

The simplest and most striking evidence in the field relationships category is the good agreement of cleaned directions from within individual geological units (89 per cent of flows with α_{95} values for mean directions of $< 5°$) and the sharp changes in direction at a number of horizons in the lava pile (for example; R → shallow R → R for C079 → C081 to 88 → C089/90 and C112/3 → C114 → C116). Again, three of four dykes which are altered in the same manner as the lavas they intrude, have narrow lava contact zones with cleaned NRM directions close to dyke directions (Fig. 2). Both N and R dykes occur in this set. The fourth dyke, and its contact rock, were both extremely soft magnetically and no clear stable directions were obtained. In a similar baked-rock test involving a small set of thin intrusive sheets no matching of sheet and adjacent lava directions was observed (the explanation of this latter result may be in very restricted baked zones (see, e.g. Ade-Hall & Dagley 1974)).

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A further observation in this category is that the average cleaned directions are statistically identical for the sets of lavas that have experienced, respectively, laumontite and the more extreme prehnite-epidote alteration conditions. In addition both sets contain flows with shallow reverse directions; these are found elsewhere in the British Tertiary province, in some instances in rocks that have experienced only low (zone B, 100°C) hydrothermal alteration conditions (Ade-Hall *et al.* 1972; Wilson, Dagley & Ade-Hall 1972).

Remanence analyses, by the alternating field method, showed that NRMs consisted of not more than two components, the more stable one generally showing good within unit directional consistency. Only the thermal method allows complete demagnetization of NRM. By this method the NRM of sample C112.14, from the zone of higher alteration, was seen to be composed of two components. The lower temperature component was removed by 270°C leaving a well defined reverse component that persisted until the sample moment approached the random noise level at over 560°C.

Other results that are relevant here are the observations by Wilson & Smith (1968), Marshall & Cox (1971) and Ryall & Ade-Hall (1975) that magnetization *direction* persists through several types of magnetic mineral alteration processes. The results of a number of field studies have similar implications (Ade-Hall *et al.* 1971, 1972; Ade-Hall, Johnson & Ryall 1976) while other studies are taken as showing that directional change has taken place or may take place during alteration (Storetvedt 1968; Storetvedt & Peterson 1970; Palmer 1970; Lovlie, Gidskehaug & Storetvedt 1972; Johnson & Merrill 1973; Ade-Hall *et al.* 1973; Lovlie & Kvingedal 1975). No single picture of how basalt magnetization behaves during alteration can be drawn from these varied results. Some factors which appear to need systematic investigation are the type and intensity of alteration and the geometrical relationship of TRM and field during alteration. The experimental difficulty in identifying with certainty small high temperature components of magnetization (e.g. Lovelie *et al.* 1972, p. 492) and ambiguities in the interpretation of alteration products in terms of thermal histories have contributed to the variety of interpretations. The results of two earlier studies of the Mull lavas (Ade-Hall 1964; Wilson 1964) may be interpreted in terms of the occurrence of extensive directional remagnetization. These studies show that reverse polarities are strongly associated with high deuteric oxidation. However, since the alteration in question took place during initial cooling, and stable directions were not isolated, the correlation seems most readily explained by the relationship between NRM stability and deuteric oxidation state (Strangway, Larson & Goldstein 1968; Ade-Hall 1969).

In summary, the most convincing evidence on the question of directional remagnetization is of the field relationship type. Here all the clear results favour the identification of original TRM directions with stable NRM directions.

6 Analysis of remanence directions

The fundamental assumption in the application of paleomagnetism to crustal geology has been that if sufficient units from a lava pile are sampled, the overall average direction will coincide closely with the time-average dipole field inclination in the sampling area at the time of magnetization of the rock units. However, as remanence cleaning methods have been perfected, with increase in the precision with which remanence directions can be identified, it has become clear both that the geomagnetic field can be quite irregular in its detailed behaviour (Heinricks 1967; Lawley 1970; Wilson, Dagley & McCormack 1972 and others) and that lava extrusion frequency, and thus magnetic-field sampling rate, varies widely (Watkins & Gunn 1969; Wilson 1970; Doell 1972 and others). The problem is one of identifying contributions to the data from the time-averaging dipole field, excursions from

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the dipole field and variations in the rate of lava extrusion, and then separating the different effects. As a first step, the minimum number of temporally independent geomagnetic field values in the data is determined (Table 2). This is done by examining the data for sequences of closely matched NRM directions. The assumption is then made that these groups represent periods of rapid lava extrusion compared with the rate of secular change at the field. As such, these sequences only contain a single direction of the magnetic field. The recent behaviour of several volcanoes (St Helens, Fuji and Vesuvius), consisting of bursts of activity separated by quiescent periods of up to 500 yr, supports this interpretation of the paleomagnetic data (Crandell, Mullineaux & Rubin 1975). The recognition criterion for such sequences is that α_{95} limits for successive flows overlap, but exceptions are made where α_{95} values exceed 5° or when a single relatively badly-defined NRM direction is interposed between two well-defined, statistically identical directions. Many short sequences of directions satisfy these criteria and 22 such sequences, containing 83 of the 139 flow mean directions, are identified. Individual sequences consist of from 2 to 12 flows. This means of identifying sequences ensures that groups of flow units (such as units 53A/B/C/D, which were identified as such in the field), flows of different source or chemistry but close temporal relationships, together with 'flows' from areas of incomplete exposure that are in fact parts of the same flow, are all recognized. This exercise is satisfactory for the purpose of obtaining the true number of geomagnetic field directions recorded if only variations in lava-eruption frequency has occurred, but it will yield a minimum number of directions if evidence for long intervals of low secular variation being suppressed. For the purpose of this analysis we find average directions for each of these 22 sequences (Table 2) which are then given unit weight, as are the directions for the remaining 56 single flows, in calculating an overall average direction. This treatment might have to be modified at a later date if some means of separating the effects of secular variation and extrusion rate becomes available.

7 Implications of the paleomagnetic directions from the Mull lavas

7.1 POLARITY OF THE LAVA SEQUENCE

The flows are all reversely or shallow reversely magnetized, although about 20 per cent of the dykes of the North Mull dyke swarm (Ade-Hall *et al.* 1972) and the North Skye dyke swarm (Wilson *et al.* 1974) are normally magnetized.

The Mull reversed sequence contains significantly more flows than the longest of the 61 Icelandic normal and reverse sequences described in Dagley *et al.* 1967. The greater number of flows in the Mull sequence may reflect the proximity of the sections to the Mull volcanic centre, the separation being at the most a few kilometres for the uppermost part of the section. However, this result is also consistent with the greater average length of polarity intervals in the Paleogene (Heirtzler *et al.* 1968; Cox 1969a).

7.2 OTHER RESULTS CONCERNING THE GEOMAGNETIC FIELD

The distribution of temporally independent directions of Fig. 3 is axisymmetric and may be treated as a Fisherian distribution for the purpose of obtaining average direction and poles. Wilson *et al.* (1972) have shown, for a large collection of Icelandic Neogene basalts, that the stable N and R states of the geomagnetic field are limited, in terms of pole positions, to within 40° of the geographic pole. If the Icelandic result is applicable here, and if we assume that the average direction of the cluster of directions of Fig. 3 represents a pole coincident with the geographic pole during the eruptive activity, then all the directions have poles within the limits of the stable states of the field. The average direction is then $-57.5°$ at 182.1° E, k : 32, α_{95} : 2.8 (Table 3).

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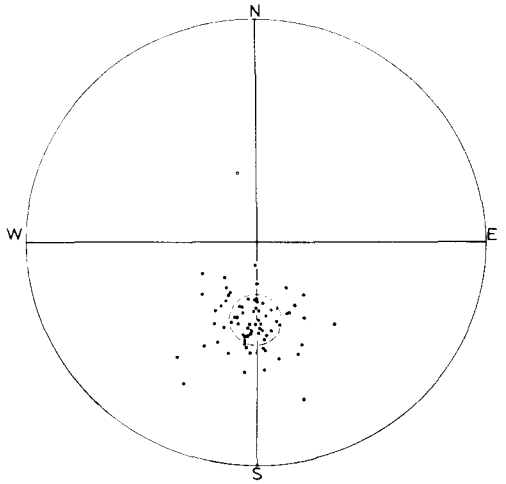


Figure 3. Equal angle stereoprojection showing the distribution of the 78 temporally independent paleomagnetic directions for the Mull lavas.

If the data are only subdivided into a few large subgroups the time required for accumulation of the data in each subgroup may be expected to be closely related to the number of directions in the subgroup. On this basis, the field direction was within 10° of the mean direction (ringed on Fig. 3) for 52 per cent of the time of eruption of the lava sequence. The field direction was between 10° and 20° of the mean direction for 30 per cent of the time of eruption of the lava sequence, and between 20° and 40° of the mean direction for the remaining 18 per cent of the time.

A conventional means of describing the distribution of directions in Fig. 3 is by a secular variation estimate. Following Doell (1970) we calculate as a measure of secular variation the angular dispersion of virtual geomagnetic poles (VGP's) using the approximation

$$S_F^2 \simeq 2/K_F$$

where K_F is the dispersion of VGP's with experimental errors and the effects of local anomalies removed. The value of S_F obtained is $17.6^{+2.2}_{-1.7}{}^{\circ}$ – the errors quoted are at the 95 per cent confidence limits and have been estimated by the method of Cox (1969b). This value is consistent with the predictions for secular variation versus latitude for a model in which a non-dipole field of amplitude invariant with latitude is added to a fixed dipole. The lower 95 per cent confidence limit is also close to values predicted for other combinations of dipole and non-dipole fields (Doell 1970) and only differs clearly from a model based on dipole wobble alone. If the results for Hawaii (S_F = 10.8°) represent dipole wobble alone then the non-dipole field contribution in Mull over the interval of lava eruption was 13.9°, or about 30 per cent larger than the dipole wobble contribution. This estimate may be excessive if evidence for periods of low secular variation has been suppressed by the averaging process.

7.3 SIGNIFICANCE FOR CRUSTAL GEOLOGY

In addition to mapping the polarity succession and secular variation behaviour of part of the Paleogene geomagnetic field, the precise paleomagnetic evidence now becoming available from the British Tertiary igneous province should also constrain the possible motions of either or both the geomagnetic source or the Eurasian Plate. Since this is the last of a series

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Table 3. Mean directions and pole positions for the Mull lavas.

	N	$D(^\circ)$	$I(^\circ)$	k	$\alpha_{95}(^\circ)$	$\lambda(^\circ)$	$\phi(^\circ)$	$\delta_p(^\circ)$	$\delta_n(^\circ)$
Mull lavas: all temporally independent directions	78	182.1	−57.5	32	2.8	71.2 N	168.7 E	3.0	4.1
Mull lavas: temporally independent laumontite zone directions	25	182.1	−59.7	19	6.4	74.0	168.3	7.3	9.7
Mull lavas: temporally, independent epidote zone directions	53	182.2	−56.0	47	2.8	70.0	168.8	2.9	4.0
Mull lavas: mean pole (Lat: 56.4° N Long: 353.9° E)	78					72.2	168.3	k 21	$\alpha_{95}(^\circ)$ 3.4
Garron Point Antrim lavas (Wilson, 1970) mean pole (Lat: 55.1° N Long: 354.0° E)	25					70.9	163.9	25	5.6
Combined Mull and Garron Point lavas, mean pole	103					71.9	167.2	22	3.0

of three papers describing lava paleomagnetism in the province the collected results are summarized here. Several extensive studies of dyke swarm in the province have still to be completed but since time sequence is generally unknown for dyke swarms the paleomagnetic information is less useful than that from lava sequences.

Of the recent studies of lava sequences in the province only the Garron Point (Northern Ireland) basalts yield an approximately Fisherian distribution of directions (Wilson 1970). The Skye lavas (Wilson *et al.* 1972) yield a distribution of directions that shows distinct azimuthal elongation which many imply unrecognized non-dipolar behaviour of the field. The mean pole positions for the Garron Point and Skye lavas are shown in Fig. 4. These well defined pole positions are not significantly different at the 95 per cent level. For this reason it seems justifiable to average the Mull and Garron Point poles to give a best pole position for the British Tertiary igneous province, in spite of there being some uncertainty about the age relationships of the two lava sequences. Thus, Purdy *et al.* (1972) suggest a *minimum* age of 49.9 ± 2.2 Myr for the Garron Point lavas while Mussett *et al.* (1973) propose an age of 62 ± 0.5 Myr for the Mull lavas. In contrast, MacIntyre (1973) and Evans, Fitch & Miller (1973) find a closer grouping of ages for, respectively, the central intrusions of all the igneous centres and the volcanic sequences of the province. A weighted mean pole for the Mull and Garron Point lavas is at 71.9° N, 167.2° E (k : 22, α_{95} : 3.0°) with a corresponding paleolatitude for the area of 37° N. This pole should be accepted as the best value for Northern Britain at about 55 Myr. It should replace similar but less rigorously obtained values summarized in Irving (1964, poles 11.012–11.027) and others postdating Irving's compilation (Smith 1966; Dagley 1969; Ade-Hall *et al.* 1972; Lovlie *et al.* 1972). The important feature of this pole position is the large difference between the paleolatitude of 37° N for the lavas and the present latitude of 56° N.

At least four mechanisms could account for this latitude difference: (i) Polar wander. (ii) Northwards displacement of the axial dipole. (iii) A strong local, non-dipolar field. (iv) Crustal motion.

Polar wander would require a net polar motion of some 18° latitude, approximately along

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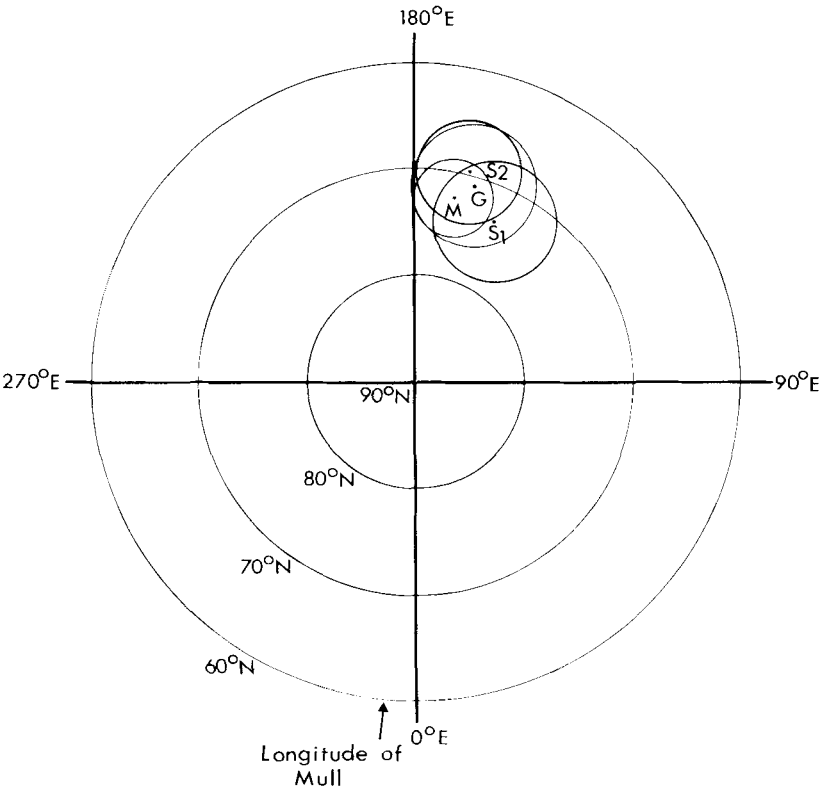


Figure 4. Average pole positions for the Mull, Antrim and Skye lava sections (M: Mull, G: Garron Point, Antrim, S1: Skye, not tilt corrected, S2: Skye, tilt corrected, circles indicate α_{95} values).

longitude 180° E, during the last 55 Myr. If such motion had occurred an apparent southwards relative motion of the Earth's crust in the Pacific would be seen. In fact the opposite effect is observed: Francheteau *et al.* (1970) and Vaquier & Harrison (1972) find from an analysis of Pacific seamounts magnetization that the North-western Pacific appears to have moved northwards by 35° of latitude over the last 90 Myr. Thus, polar wander can only be an effective mechanism if northwards plate motion in the Pacific during the Cenozoic considerably exceeded oppositely directed polar wander.

Wilson (1970, 1971) has shown that global Late Tertiary, Quaternary and Recent palaeomagnetic results can be modelled by a time-average northwards displacement of the axial dipole. The average displacement was 285 ± 74 km during the Upper Tertiary. This displacement is seen as shallower northern hemisphere field inclinations than would be produced by a centred axial dipole. Since this is the relationship between remanence inclination and geographic latitude for the Mull lavas we must consider whether Wilson's model can reasonably explain our results. Close to 1400 km northwards displacement is required to produce our result. While a displacement of this magnitude still leaves the source well within the core, it is more than four times greater than the Upper Tertiary average value. If correct, this displacement represents extreme dipole offset with possible implications for dynamo models.

A persistent local non-dipole field anomaly could have been responsible for the 18° difference between paleomagnetic and geographic latitudes. Such an anomaly could only be identified with certainty if rocks in the adjacent Western Europe and North Atlantic

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areas of the same age as the Mull lavas yielded well defined paleolatitudes close to present geographic latitudes. In fact, examination of the poles for Paleogene sampling sites from Greenland and the Faeroes to Southern Europe (McElhinny 1973) shows that the British Tertiary igneous province poles are not exceptional in the set. All but one of the 14 poles considered show latitude offset from the geographic pole of from 14° to 25°. The only clear division in the set is between western poles (Greenland, Faeroes, Britain, Portugal and Italy) with longitudes of from 160° to 200° E and eastern poles (France, Germany, Poland, Bulgaria and Italy) with longitudes of from 105° to 145° E.

In terms of the absolute plate motion, the 18.1° offset of the average paleomagnetic pole from the geographic pole implies a net northwards motion of the Eurasian Plate of 2010 km in the last 55 Myr at an average rate of 3.7 cm/yr. Similar motions are found for the American and Pacific Plates, and from the older parts of the European paleomagnetic record (McElhinny 1973; Francheteau *et al.* 1970; Vaquier & Harrison 1972). An apparent convergence of plates towards the Arctic is implied, which may have been accommodated by compressional folding in the Verkhoyansk Mountains and Sverdrup Basin, and Brooks Range, and by subduction in Eastern Siberia and the Bower Ridge (Pitman & Talwani 1972). Since strike-slip motion along the axis of the Mid-Atlantic Ridge is unknown, it seems that several plates may have a relatively large common component of motion orthogonal to motion at intervening plate margins. This model requires very different directions of flow at different levels in the mantle. The fact that paleomagnetic and plume expression (Morgan 1972b) absolute motion estimates for the Eurasian Plate are rather different is of interest and may imply a limited depth extent for plumes.

Conclusion

Two major results derive from this study: the first concerns the important question of whether a high degree of hydrothermal alteration of basalts, with associated changes in the opaque minerals, is accompanied by directional remagnetization. The evidence suggests that extensive directional remagnetization has not taken place here.

The second result is that the preferred mean pole position for the British Tertiary igneous province lavas at 71.9° N, 167.2° E, with α_{95} value of 3.0° is well separated from the geographic pole. Both geomagnetic and crustal motion effects may contribute to the differences between the paleomagnetic and geographic poles. If the paleomagnetic pole results from northwards plate motion since the early Tertiary then current views of plate motions may have to be modified to include the motion of coupled plates in a direction approximately orthogonal to motion at intervening margins. This situation would require more complicated upper mantle motions than are usually envisioned. Apparent northwards motion of the Pacific, American and Eurasian plates during the Tertiary focuses attention on our need for complete knowledge of the geological structures of the Arctic Basin and its environs.

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